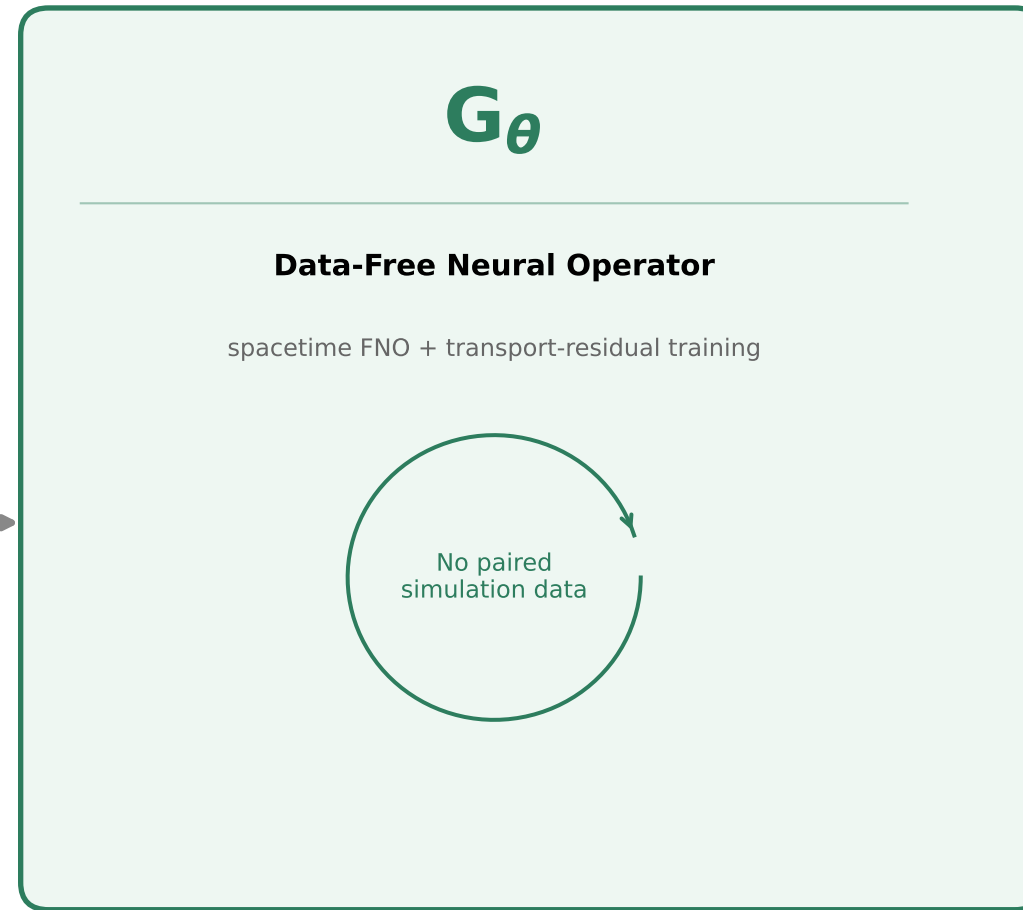
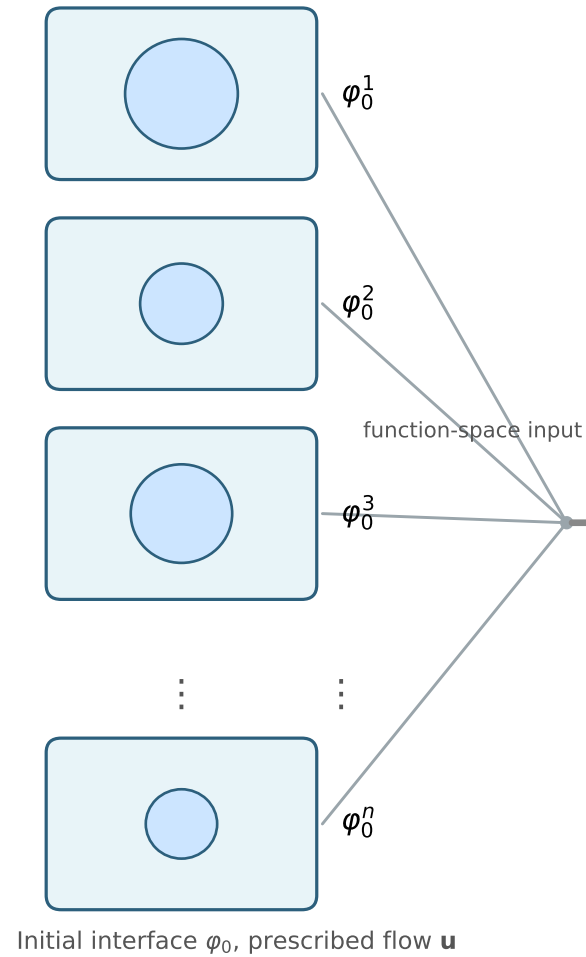
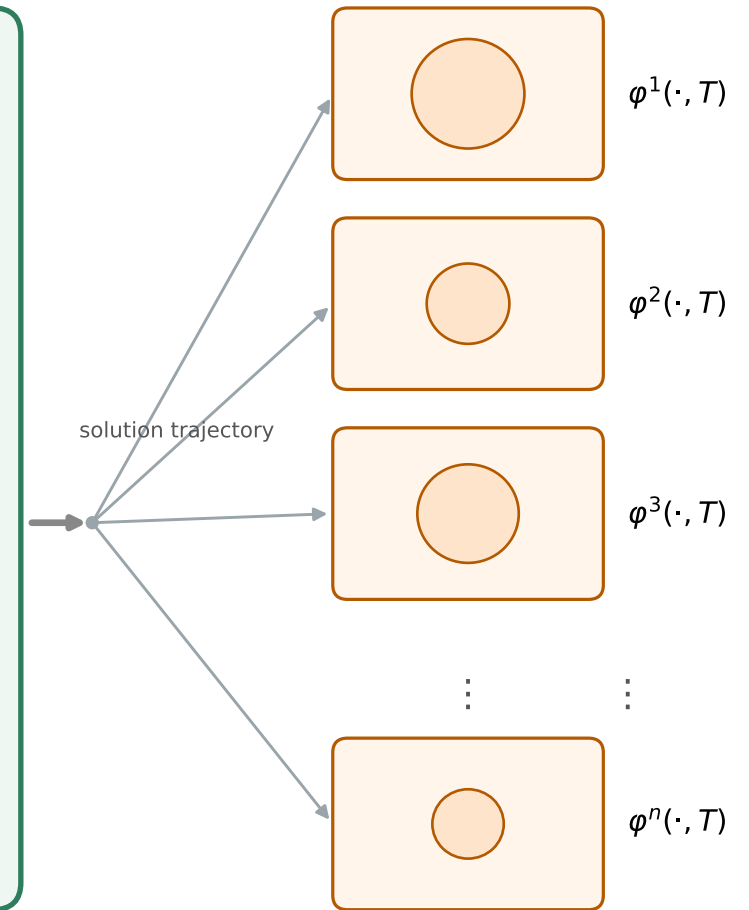


Data-Free Neural Operator for Level-Set Interface Advection

Input Function Space



Output Function Space



Residual loss $\mathcal{L}(\theta) = \mathcal{L}_{\text{pde}} + w_{\text{eik}} \mathcal{L}_{\text{eik}}$
no reference solutions needed

✓ Generalises across initial conditions

✓ Data-free training

✓ Interface is the PDE unknown